

Hyeongjun Yang

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RESEARCH INTERESTS

Applications: *AI Agents / Task-oriented Dialogue Systems / Contextual Retrieval*

Approaches: *Multimodal Knowledge Augmentation / Data-efficient Learning / LLM Distillation / PEFT*

WORK EXPERIENCE

Samsung Electronics, South Korea
Staff Engineer, Language AI Team

Mar 2026 –

Yonsei University, South Korea
Graduate Researcher, ICL Lab

Mar 2020 – Feb 2026

EDUCATION

Yonsei University, South Korea
Ph.D. in Computer Science (Advisor: Prof. Kyong-Ho Lee)
- Thesis: "*Towards Conversational Recommender Systems with Contextual Intelligence*"

Mar 2020 – Feb 2026

California State University, San Bernardino, United States
Exchange Student in Computer Science

Sep 2018 – Jun 2019

Dankook University, South Korea
B.S. in Software Engineering
- Mandatory military service (Oct 2015 - Jul 2017)

Mar 2013 – Feb 2020

PUBLICATIONS

(★ denotes first-authored publications)

[15] "Retrieval-Augmented Knowledge Graph Validation with Large Language Model Reasoning"
Donghyun Kim, Hyeongjun Yang, Chanhee Lee, Seokju Hwang, Kyong-Ho Lee
Under Review

[14] "Reliable Distillation of Recommendation Rationales from Large Language Models for Enhanced Conversational Recommendation"
Hyeongjun Yang★, Donghyun, Donghyun Kim, Kyuhwan Yeom, Kyong-Ho Lee, Byungkook Oh, Xiongnan Jin
IEEE Transactions on Knowledge and Data Engineering 2026 [paper]

[13] "Causality-Guided Spurious Evidence Pruning with GFlowNet for Faithful Graph Retrieval"
Jinho Choi, Hye-Yoon Baek, Sang-Hyun Park, Sangjune Kim, Hyeongjun Yang, Byungkook Oh
CIKM 2026

[12] "Data-Efficient Adaptation to Contextual Shifts in LLM-based Conversational Recommendation"
Hyeongjun Yang★, Kyong-Ho Lee
ACL 2026 (findings) [paper]

[11] "LLMs as Knowledge Graph Refiners: Mitigating Factual Inconsistencies in Generative Knowledge Extraction"
Donghyun Kim, Hyeongjun Yang, Seokju Hwang, Kyong-Ho Lee, Chanhee Lee
ACL 2026 [paper]

[10] "RPO-RAG: Aligning Small LLMs with Relation-aware Preference Optimization for Knowledge Graph Question Answering"

Kaehyun Um, Kyuhwan Yeom, Haerim Yang, Minyoung Choi, Hyeongjun Yang, Kyong-Ho Lee
WWW 2026 [paper]

[9] "Active Learning Framework for Improving Knowledge Graph Accuracy"
Donghyun Kim, Hyeongjun Yang, Seokju Hwang, Kyuhwan Yeom, Midan Shim, Kyong-Ho Lee
IEEE Access 2025 [paper]

[8] "CoreSense: Social Commonsense Knowledge-Aware Context Refinement for Conversational Recommender System"
Hyeongjun Yang★, Donghyun Kim, Gayeon Park, KyuHwan Yeom, Kyong-Ho Lee
IEEE Transactions on Knowledge and Data Engineering (Vol. 37) 2025 [paper]

[7] "Knowledge-constrained Interest-aware Multi-Behavior Recommendation with Behavior Pattern Identification"
Gayeon Park, Hyeongjun Yang, KyuHwan Yeom, Myeongheon Jeon, Yunjeong Ko, Byungkook Oh, Kyong-Ho Lee
Information Sciences (Vol. 692) 2025 [paper]

[6] "Granular Intents Learning via Mutual Information Maximization for Knowledge-aware Recommendation"
Hyeongjun Yang★, Yerim Lee, Gayeon Park, TaeYoung Kim, Heesun Kim, Kyong-Ho Lee, Byungkook Oh
Knowledge-based Systems (Vol. 306) 2024 [paper]

[5] "Bottom-up propagation of hierarchical dependency for multi-behavior recommendation"
{TaeYoung Kim, Hyeongjun Yang★}, Gayeon Park, Seungmi Lee, Kyong-Ho Lee
Engineering Applications of Artificial Intelligence (Vol. 138) 2024 [paper]

[4] "Embedding Two-View Knowledge Graphs with Class Inheritance and Structural Similarity"
Kyuhwan Yeom, Hyeongjun Yang, Gayeon Park, Myeongheon Jeon, Yunjeong Ko, Byungkook Oh, Kyong-Ho Lee
KDD 2024 [paper]

[3] "Cross-Modal Contrastive Learning for Aspect-based Recommendation"
Heesoo Won, Byungkook Oh, Hyeongjun Yang, Kyong-Ho Lee
Information Fusion (Vol. 99) 2023 [paper]

[2] "Confident Action Decision via Hierarchical Policy Learning for Conversational Recommendation"
Heeseon Kim, Hyeongjun Yang, Kyong-Ho Lee
WWW 2023 [paper]

[1] "CLICK: Contrastive Learning for Injecting Contextual Knowledge to Conversational Recommender System"
Hyeongjun Yang★, Heesoo Won, Youbin Ahn, Kyong-Ho Lee
EACL 2023 [paper]

PROJECTS

Industrial Projects

Agentic AI Development for Samsung Bixby Mobile Assistant Mar 2026 – Present
at Samsung Electronics

- **Project member:** Core Algorithm Design & Optimization for Agentic AI Service
- Agentic Resource Discovery Development (Contextual Retrieval for Prompt Optimization)

Academic Research Projects

LLM-based Attribute-aware Information Retrieval from Knowledge Graphs Dec 2024 – Dec 2025
w/ Samsung Research

- **Project Member:** Proposal / Research Planning

- Related Publication: [10]

Refining User Preferences by Leveraging Knowledge Graphs for Conversational Recommender System

Mar 2022 – Feb 2025

w/ National Research Foundation of Korea (NRF)

- **Project Leader:** Proposal / Research Planning / Development / Validation / Publication
- Related Publications: [1], [2], [3], [8], [9]

Query Intent-based Information Retrieval from Knowledge Graphs

Dec 2023 – Dec 2024

w/ Samsung Research

- **Project Member:** Proposal / Research Planning

Ontology-based Multi-type Interaction Modeling

Dec 2022 – Dec 2023

w/ Samsung Research

- **Project Leader:** Proposal / Research Planning / Development / Validation / Publication
- Related Publications: [4], [7]

Knowledge Graph based Intent-Community Detection

Dec 2021 – Dec 2022

w/ Samsung Research

- **Project Leader:** Proposal / Research Planning / Development / Validation / Publication
- Related Publications: [5], [6]

Development of XAI-based Technology for Smart Energy Platform

Mar 2020 – Feb 2021

w/ Korea Electric Power Research Institute (KEPRI)

- **Project Member:** Data Collection / Validation
- Energy Ontology Design & Knowledge Graph Construction

SKILLS

Programming Languages: *Python / TypeScript / C / C++*

AI & ML Frameworks: *PyTorch / HuggingFace / LangChain / TensorFlow*

Knowledge Graphs & Ontologies: *RDF / OWL / SPARQL*

DevOps: *Git / Docker / Kubernetes*